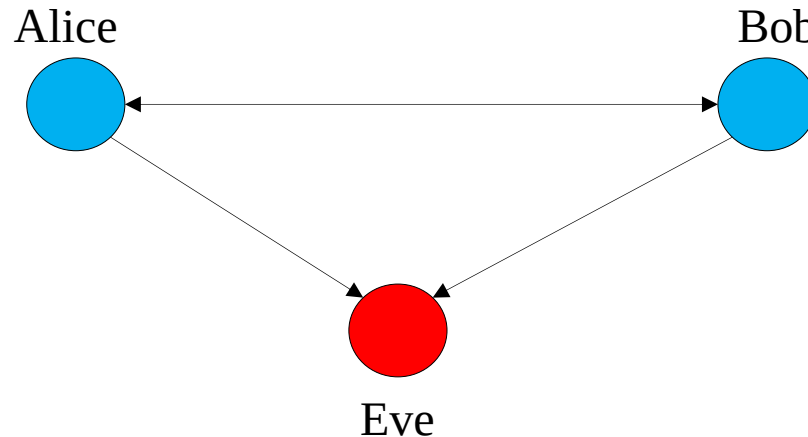


Quantum Key Distribution

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Introduction

When describing the secure exchange of messages between two users, Alice and Bob, we use the model shown below:



A third user, Eve, is an unauthorized user who is attempting to intercept the messages exchanged between Alice and Bob.

To prevent this, a key needs to be used by Alice and Bob that will encrypt their messages and make it almost impossible for Eve to break.

Encryption

- The purpose of encryption is to:
 - Prevent unauthorized users from receiving a message (secrecy)
 - Verify the source of the message (authentication)
 - Ensure the received message has not been altered (integrity)
- Deciphering a message is only possible if the intended recipient knows the deciphering algorithm and the key.

Private Keys

- One type of key is known as a **one time pad (Vernan cipher)**, which satisfies the criterion for perfect secrecy.
- It is used only once to encrypt only a single character and is then discarded.
- The key is added to the character by modulo-2 addition and then Bob can recover the character by adding the key again.

Private Keys

- Private Key Distribution suffers from the **key distribution problem**. How do we securely deliver the key to Alice and Bob without Eve obtaining it?
- Security depends on knowledge of a single key only being available to Alice and Bob. Therefore, this encryption scheme is known as a **symmetric key system**.

Public Keys

- A solution to this problem is public key distribution. Here two keys are generated: a public key and a private key, where one is used for encryption and the other is used for decryption.
- One of the most popular public key encryption algorithms is RSA (Rivest-Shamir-Adleman), where encryption uses a public key and decryption uses a private key.
- The idea of RSA is based on **Diffie-Hellman Key Exchange**

Multiplicative Groups

The Diffie-Hellman Key Exchange algorithm requires knowledge of multiplicative groups modulo- p , $\mathbb{Z}_p$.

A multiplicative group modulo- p is a set of integers $\{0, 1, 2, \dots, p - 1\}$, where p is prime. If any two element in the group are multiplied modulo- p , then the result will also be in the group.

A multiplicative group contains primitive elements, where raising the element to different power $0, 1, \dots, p - 1$ will result in another non-zero element in the group

For $p = 5$, the primitive elements are 2 and 3.

$$\begin{aligned}\mathbb{Z}_5 &= \{0, 1, 2, 3, 4\} \\ &= \{0, 2^1 \bmod 5 = 2, 2^2 \bmod 5 = 4, 2^3 \bmod 5 = 3, 2^4 \bmod 5 = 1\} \\ &= \{0, 3^1 \bmod 5 = 3, 3^2 \bmod 5 = 4, 3^3 \bmod 5 = 2, 3^4 \bmod 5 = 1\}\end{aligned}$$

Diffie-Hellman Key Exchange

- Alice and Bob publicly announce a prime number p and a primitive element g .
- Alice privately chooses a large integer a and then publicly transmits $A = g^a \bmod p$ to Bob.
- Bob privately chooses a large integer b and then publicly transmits $B = g^b \bmod p$.
- Alice privately computes the decryption key $K = B^a \bmod p$ and Bob privately computes $K = A^b \bmod p$.

The keys match because $(g^a \bmod p)^b \bmod p = (g^b \bmod p)^a \bmod p = g^{ab} \bmod p$

Eve knows g , p , A and B , but it is difficult to determine a (or b) because $g^a \bmod p = A$ has an infinite number of solutions. **This is known as the Discrete Logarithm Problem.**

Diffie Hellman Key Exchange Example

Alice and Bob publicly agree to select a prime number $p = 17$ and a primitive element $g = 2$.

Alice privately chooses a random integer $a = 19$ and then calculates

$$A = g^a \bmod p = 2^{19} \bmod 17 = 8$$

At the same time, Bob privately chooses a random integer $b = 10$ and calculates

$$B = g^b \bmod p = 2^{10} \bmod 17 = 4$$

Alice and Bob publicly exchange A and B

Diffie Hellman Key Exchange Example

Alice now calculates the key

$$K = B^a \bmod p = 4^{19} \bmod 17 = 13$$

At the same time, Bob calculates key

$$K = A^b \bmod p = 8^{10} \bmod 17 = 13$$

As expected, both keys match and are only known to Alice and Bob

Eve could only obtain this key if she can determine $g^{ab} \bmod p$, given that she knows $g^a \bmod p$ and $g^b \bmod p$. This is known as the **Diffie-Hellman Problem**.

Quantum Key Distribution

- Diffie-Hellman key exchange is secure based on the assumption that the discrete logarithm problem cannot be solved if the prime number chosen is sufficiently large.
- However, it is susceptible to the publicly transmitted information being intercepted and modified (no authentication).
- Unlike classical key distribution, quantum key distribution is provably secure.
- An eavesdropper cannot clone or gain any information from transmitted qubits without disturbing their state.

Quantum Key Distribution

If Eve tries to distinguish between two non-orthogonal quantum states, she can only gain information by introducing a disturbance to the signal.

Let $|\psi\rangle$ and $|\phi\rangle$ be two non-orthogonal quantum states that Eve wish to gain information about.

She uses an ancillary qubit $|u\rangle$ to unitarily interact with the two states that does not disturb the states.

$$U|\psi\rangle|u\rangle \rightarrow |\psi\rangle|v\rangle$$

$$U|\phi\rangle|u\rangle \rightarrow |\phi\rangle|v'\rangle$$

Eve wants $|v\rangle$ and $|v'\rangle$ to be different to gain information about the identity of the state

Quantum Key Distribution

However, inner products are preserved under unitary transformations

$$(U|\psi\rangle|u\rangle, U|\phi\rangle|u\rangle) = (|\psi\rangle|v\rangle, |\phi\rangle|v'\rangle)$$

$$\langle u|\langle\psi|U^\dagger U|\phi\rangle|u\rangle = \langle v|\langle\psi|\phi\rangle|v'\rangle$$

$$\langle\psi|\phi\rangle\langle u|u\rangle = \langle\psi|\phi\rangle\langle v|v'\rangle$$

Therefore, $\langle v|v'\rangle = \langle u|u\rangle = 1$, which mean that $|v\rangle$ and $|v'\rangle$ must be identical.

This proves that you cannot distinguish between $|\psi\rangle$ and $|\phi\rangle$ without disturbing at least one of them.

The BB84 QKD Protocol

1. Alice chooses $(4 + \delta)n$ random data bits, a .
2. Alice chooses a random $(4 + \delta)n$ -bit string b . Each data bit is encoded as $\{|0\rangle, |1\rangle\}$ if $b = 0$ or $\{|+\rangle, |-\rangle\}$ if $b = 1$.
3. Alice transmits her states to Bob on the quantum channel.
4. Bob choose a random $(4 + \delta)n$ -bit string b' . If $b' = 0$, Bob measures the received state in the Z basis. If $b' = 1$, Bob measures the state in the X basis.
5. Alice announces b on the public channel.
6. Alice and Bob discard any bits where Bob measured a different basis than Alice prepared.

The BB84 QKD Protocol Example

Alice's random bits, a	Alice's random bits, b	Alice's quantum states	Bob's random bits, b'	Bob's basis	Bob's measurement of Alice's quantum states to obtain a'
1	0	$ 1\rangle$	1	X	$H 1\rangle \rightarrow -\rangle$: 0 or 1
0	1	$ +\rangle$	1	X	$H +\rangle \rightarrow 0\rangle$: 0
1	1	$ -\rangle$	0	Z	$I -\rangle \rightarrow -\rangle$: 0 or 1
1	0	$ 1\rangle$	0	Z	$I 1\rangle \rightarrow 1\rangle$: 1
0	1	$ +\rangle$	1	X	$H +\rangle \rightarrow 0\rangle$: 0
0	0	$ 0\rangle$	0	Z	$I 0\rangle \rightarrow 0\rangle$: 0
1	0	$ 1\rangle$	1	X	$H 1\rangle \rightarrow -\rangle$: 0 or 1
0	1	$ +\rangle$	0	Z	$I +\rangle \rightarrow +\rangle$: 0 or 1

The BB84 QKD Protocol Example

- Notice that when $b \neq b'$, a' is equally likely to be 0 or 1 (uncorrelated with a), but when $b = b'$, $a' = a$.
- Assume Alice and Bob keep $2n$ bits of their result. Alice selects n bits at random and publicly announces their positions.
- Alice and Bob then announce and compare the bit values. If more than t bits disagree, then they stop and restart the protocol.

Effect of Eve on BB84 QKD Protocol

Now, let us assume that Eve intercepts Alice's quantum states before Bob

Eve receives: $|1\rangle|+\rangle|-\rangle|1\rangle|+\rangle|0\rangle|1\rangle|+\rangle$ and chooses a random bit sequence $b'' = 1\ 0\ 1\ 0\ 0\ 1\ 0\ 1$ to determine the basis she measures Alice's quantum states in

Using the basis XZXZZXZX, Eve obtain the quantum states:

$$|-\rangle|+\rangle|1\rangle|1\rangle|+\rangle|+\rangle|1\rangle|0\rangle.$$

Bob will now receive Eve's quantum states: $|-\rangle|+\rangle|1\rangle|1\rangle|+\rangle|+\rangle|1\rangle|0\rangle$.

Effect of Eve on BB84 QKD Protocol

Using the same basis that Bob generated in the previous example, Bob applies this to the received quantum states and obtains:

$$|1\rangle|0\rangle|1\rangle|1\rangle|0\rangle|+\rangle|-\rangle|0\rangle$$

Measuring these quantum states will result in the binary sequence $a' = 10110\underline{11}0$ where the 6th and 7th bit were equally likely to be a 0 or 1 and a bit value of 1 was the result for both bits.

Alice announces b on the public channel and Alice and Bob discard any bits where Bob measured a different basis. In this example, Alice and Bob compare bits 2 and 6.

Due to Eve, Bob's bits are 01 and Alice's bits are 00 so the key has been corrupted.

The B92 Protocol

1. Alice prepares a random bit a and sends Bob the state $|0\rangle$ if $a = 0$ or $|+\rangle$ if $a = 1$.
2. Bob prepares a random bit a' and measures the qubit in the Z basis $|0\rangle, |1\rangle$ if $a' = 0$, or in the X basis $|\pm\rangle$ if $a' = 1$.
3. Bob obtains the bit b and announces b on the public channel.
4. Alice and Bob keep only those pairs $\{a, a'\}$ where $b = 1$.
5. The final key is a for Alice and $\bar{a}'$ (the complement of a') for Bob.

The B92 QKD Protocol Example

Alice's random bits, a	Alice's quantum states	Bob's random bits, a'	Bob's basis	Bob's measurement of Alice's quantum states to obtain b
1	$ +\rangle$	1	X	$H +\rangle = 0\rangle$: 0
0	$ 0\rangle$	1	X	$H 0\rangle = +\rangle$: 0 or 1
1	$ +\rangle$	0	Z	$I +\rangle = +\rangle$: 0 or 1
1	$ +\rangle$	0	Z	$I +\rangle = +\rangle$: 0 or 1
0	$ 0\rangle$	1	X	$H 0\rangle = +\rangle$: 0 or 1
0	$ 0\rangle$	0	Z	$I 0\rangle = 0\rangle$: 0
1	$ +\rangle$	1	X	$H +\rangle = 0\rangle$: 0
0	$ 0\rangle$	0	Z	$I 0\rangle = 0\rangle$: 0

The B92 QKD Protocol Example

Notice that when $a = a'$, b is always 0. When $a \neq a'$ then there is a 50% chance that the measured bit b is 1.

Let us assume that Bob's measured bits are: $b = 01011000$

Bob announces b and then Alice and Bob keep only those pairs (a, a') where $b = 1$.

This corresponds to the 2nd, 4th and 5th bits of b , with $a = 010$ and $a' = 101$.

Therefore, the final key for Alice is $K = a = 010$.

The final key for Bob is $K = \overline{a'} = 010$.

Effect of Eve on B92 QKD Protocol

As before, we assume that Eve has intercepted Alice's quantum states.

Eve receives $|+\rangle|0\rangle|+\rangle|+\rangle|0\rangle|0\rangle|+\rangle|0\rangle$ and chooses a random bit sequence $a'' = 01101011$ to determine the basis she measures Alice's quantum states in

Using the basis ZXXZXZXX, Eve obtain the quantum states:

$$|+\rangle|+\rangle|0\rangle|+\rangle|+\rangle|0\rangle|0\rangle|+\rangle$$

Measuring these quantum states will result in the binary sequence: ??0??00?, where ? means the measured bit is equally likely to be 0 or 1.

Effect of Eve on B92 QKD Protocol

Bob will now receive Eve's quantum states: $|+\rangle|+\rangle|0\rangle|+\rangle|+\rangle|0\rangle|0\rangle|+\rangle$.

Using the same basis that Bob generated in the table, Bob applies this to the received quantum states and obtains:

$$|0\rangle|0\rangle|0\rangle|+\rangle|0\rangle|0\rangle|+\rangle|+\rangle$$

Measuring these quantum states will result in the binary sequence $b = 000?00??$

Let us assume that Bob's measured bits are $b = 000\mathbf{1}00\mathbf{11}$

Effect of Eve on B92 QKD Protocol

Bob announces b and then Alice and Bob keep only those pairs (a, a') where $b = 1$.

This corresponds to the 4th, 7th and 8th bits of b , with $a = 110$ and $a' = 010$.

We can see that Eve's measurement of Alice's quantum states has affected Alice's and Bob's secret key, i.e $a \neq \bar{a}'$

In summary, we have seen two effective quantum key distribution protocols.

- BB84 uses 4 quantum states that are not all mutually orthogonal so no measurement can distinguish between all of them with certainty.
- B92 uses only 2 non-orthogonal quantum states. Being unable to distinguish between non-orthogonal states is the key to quantum cryptography.

The EPR Protocol

- In BB84 and B92, the key has been generated by Alice.
- However, we can generate a random key by using the properties of entanglement.
- This is achieved by preparing a set of EPR pairs in the state:

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}$$

The EPR Protocol

- The EPR pairs could be prepared by Alice, who sends half of each pair to Bob.
- Alternatively, a third party could prepare them and send each half to Alice and Bob.
- When Alice or Bob measure their qubit in the same basis, the binary bits a and a' that form the key will be correlated.
- However, we do not know a (or a') until the qubits are measured. So, the key is random.